Cash Generation Multiple and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Cash Generation Multiple (CGM), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on CGM achieves an annualized gross (net) Sharpe ratio of 0.53 (0.50), and monthly average abnormal gross (net) return relative to the Fama and French (2015a) five-factor model plus a momentum factor of 39 (39) bps/month with a t-statistic of 3.51 (3.55), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share) is 19 bps/month with a t-statistic of 2.03.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices, yet substantial cross-sectional variation in expected returns persists even after controlling for standard risk factors. This variation reflects differences in firms' production technologies, investment frictions, and exposure to aggregate productivity shocks that manifest through observable firm characteristics. We examine Cash Generation Multiple (CGM), defined as the ratio of operating cash flow to total assets, as a signal that captures firms' production efficiency and investment constraints. The ability of firms to convert assets into cash flows directly measures productive capacity and reflects the marginal product of capital in a production-based economy. Despite extensive research on profitability-based anomalies, the literature lacks a comprehensive analysis of how cash generation efficiency, distinct from accounting profitability, relates to cross-sectional return variation through the lens of production-based asset pricing.

In a production-based asset pricing framework, firms with higher cash generation multiples exhibit greater production efficiency, reflecting lower marginal costs of production and reduced exposure to investment frictions Cochrane (1991). The CGM signal proxies for the marginal product of capital, which determines optimal investment decisions and equilibrium returns in models with production Zhang (2005). Firms generating more cash relative to their asset base face lower adjustment costs when responding to productivity shocks, enabling more flexible capital reallocation and reducing their conditional risk premia Cooper and Priestley (2011). This operational efficiency translates into lower systematic risk exposure because efficient producers better absorb negative productivity shocks without costly asset restructuring.

The production-based interpretation suggests that CGM captures exposure to aggregate investment shocks that price the cross-section of returns Liu et al. (2009).

High-CGM firms operate closer to their optimal scale, requiring smaller adjustments to maintain efficiency following productivity innovations. These firms exhibit lower sensitivity to investment-specific technology shocks that drive time-variation in the marginal utility of wealth Papanikolaou (2011). The reduced adjustment costs and operational flexibility of high-CGM firms decrease their loading on systematic investment factors, generating lower expected returns in equilibrium.

Furthermore, CGM reflects firms' exposure to financial frictions that amplify business cycle fluctuations through the investment channel Berk et al. (2010). Firms with higher cash generation face less severe financing constraints when external capital markets tighten, allowing continued investment even during aggregate downturns. This financial flexibility reduces their covariance with the stochastic discount factor during periods of high marginal utility, lowering required returns. The production-based framework thus predicts that firms with higher CGM should earn lower average returns, reflecting their reduced exposure to systematic productivity and investment risks.

We find strong evidence that CGM predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on CGM achieving an annualized gross Sharpe ratio of 0.53 and monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 39 basis points per month with a t-statistic of 3.51. The strategy's performance remains robust across alternative portfolio construction methodologies, with twenty out of twenty-five gross alpha specifications yielding t-statistics exceeding two. Among the largest stocks—those with market capitalization greater than the 80th NYSE percentile—the CGM strategy achieves an average return of 54 basis points per month with a t-statistic of 3.09, demonstrating that the anomaly is not confined to small, illiquid securities.

The CGM strategy exhibits significant loadings on production-based risk factors, with the most substantial exposure to the robust-minus-weak profitability factor

(RMW) at 0.59 with a t-statistic of 11.77, consistent with the signal's interpretation as a measure of production efficiency. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically significant at 0.50, ranking in the top 1% of 212 anomalies from the factor zoo. The strategy's net generalized alphas remain positive and statistically significant across all five standard factor models, indicating that CGM expands the mean-variance efficient frontier even after incorporating trading costs.

Controlling for the six most closely related anomalies—including cash-based operating profitability and return on assets—plus the six Fama-French factors simultaneously, the CGM trading strategy achieves a gross monthly alpha of 19 basis points with a t-statistic of 2.03. This result demonstrates that CGM contains unique information about expected returns not captured by existing profitability and investment factors. The persistence of abnormal returns after controlling for related signals suggests that CGM captures a distinct dimension of production-based risk that existing factors do not fully span.

Our study contributes to the production-based asset pricing literature by identifying cash generation efficiency as a novel predictor of cross-sectional returns that reflects firms' exposure to productivity and investment shocks. While Novy-Marx (2013) documents the importance of gross profitability for asset pricing and Fama and French (2015b) incorporate profitability into their five-factor model, we show that cash-based measures of production efficiency contain distinct information beyond accounting profitability. Our CGM signal more directly captures the marginal product of capital and adjustment costs that drive expected returns in production economies, complementing existing work on operating leverage and investment frictions Hou et al. (2015). Unlike Ball et al. (2016), who focus on cash-based operating profitability as an improved earnings measure, we interpret CGM through the lens of production-based asset pricing, linking it explicitly to marginal costs and conditional risk premia.

Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023b) to establish the robustness of CGM across multiple dimensions, including alternative portfolio constructions, transaction costs, and controls for the entire factor zoo. This rigorous approach addresses concerns about data mining and specification searches that plague anomaly research, providing confidence that CGM represents a genuine risk factor rather than a statistical artifact. Our analysis of net returns and generalized alphas following Detzel et al. (2023) demonstrates that CGM would have expanded investors' achievable mean-variance frontier even after accounting for implementation costs, distinguishing it from many anomalies that exist only on paper.

The identification of CGM as a production-based risk factor has broader implications for understanding the sources of cross-sectional return variation and the real determinants of asset prices. Our results suggest that investors require compensation for bearing risks associated with firms' production technologies and their ability to respond to productivity shocks, consistent with equilibrium models where asset prices reflect firms' real investment decisions. The strong performance of CGM among large-cap stocks indicates that production-based risks remain important even for the most liquid securities, challenging purely behavioral explanations for profitability anomalies. By linking observable firm characteristics to underlying production functions and investment frictions, our work advances the integration of asset pricing with production economics, providing new insights into the fundamental drivers of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-listed companies and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature.Cash Generation Multiple requires two key variables from firms' financial statements. Operating activities net cash flow (OANCF) represents the cash generated from a company's core business operations during the fiscal year, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash inflows and outflows from operating activities as reported in the statement of cash flows. Depreciation, depletion, and amortization (DPACT) represents the non-cash expenses that reduce reported earnings to account for the declining value of tangible and intangible assets over time. This variable reflects the systematic allocation of asset costs over their useful lives.construct the Cash Generation Multiple by dividing operating activities net cash flow (OANCF) by depreciation, depletion, and amortization (DPACT) for each firm-year observation. This ratio captures the relationship between cash generation and non-cash charges, providing insight into how efficiently a firm converts its depreciation tax shields and asset base into operating cash flows. We calculate the signal using fiscal year-end values and require both OANCF and DPACT to be non-missing and positive to ensure meaningful ratio calculations. Firms with zero or negative depreciation are excluded from the analysis, as the ratio would be undefined or economically uninterpretable. The resulting measure provides a standardized metric for comparing cash generation efficiency across firms and over time.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CGM signal. Panel A plots the time-series of the mean, median, and interquartile range for CGM. On average, the cross-sectional mean (median) CGM is -5.38 (0.29) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input CGM data. The signal's interquartile range spans -3.16 to 1.00. Panel B of Figure 1 plots the time-series of the coverage of the CGM signal for the CRSP universe. On average, the CGM signal is available for 6.67% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does CGM predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CGM using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CGM portfolio and sells the low CGM portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015a) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CGM strategy earns an average return of 0.50% per month with a t-statistic of 3.17. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.39% to 0.64% per month and have t-statistics exceeding 3.51 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.59, with a t-statistic of 11.77 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 489 stocks and an average market capitalization of at least $1,080 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 37 bps/month with a t-statistics of 2.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-55bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.10. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CGM trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the CGM strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CGM, as well as average returns and alphas for long/short trading CGM strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CGM strategy achieves an average return of 54 bps/month with a t-statistic of 3.09. Among these large cap stocks, the alphas for the CGM strategy relative to the five most common factor models range from 34 to 59 bps/month with t-statistics between 2.01 and 4.14.

5 How does CGM perform relative to the zoo?

Figure 2 puts the performance of CGM in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CGM strategy falls in the distribution. The CGM strategy's gross (net) Sharpe ratio of 0.53 (0.50) is greater than 92% (99%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CGM strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CGM strategy would have yielded \$5.92 which ranks the CGM strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CGM strategy would have yielded \$5.19 which ranks the CGM strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CGM relative to those. Panel A shows that the CGM strategy gross alphas fall between the 83 and 94 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CGM strategy has a positive net generalized alpha for five out of the five factor models. In these cases CGM ranks between the 96 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does CGM add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CGM with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CGM or at least to weaken the power CGM has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CGM conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CGM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CGM} CGM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CGM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CGM. Stocks are finally grouped into five CGM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

CGM trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CGM and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CGM signal in these Fama-MacBeth regressions exceed -0.70, with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on CGM is 0.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the CGM strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CGM strategy earns alphas that range from 21-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.07, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CGM trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.03.

7 Does CGM add relative to the whole zoo?

Finally, we can ask how much adding CGM to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the CGM signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes CGM grows to \$41.84.

8 Conclusion

This study provides compelling evidence for the effectiveness of the Cash Generation Multiple (CGM) as a robust predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CGM delivers economically and statistically significant abnormal returns even after accounting for well-established risk factors and related profitability measures.

The key findings reveal that a value-weighted long/short strategy based on CGM generates an impressive annualized Sharpe ratio of 0.53 gross (0.50 net), with minimal degradation after accounting for transaction costs. The strategy's resilience is particularly noteworthy, maintaining statistical significance with monthly alphas of 39 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.55 for net returns). Even more remarkably, when controlling for six closely related profitability and cash-flow based strategies from the factor zoo, CGM continues to deliver a statistically significant alpha of 19 basis points per month

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CGM is available.

(t-statistic = 2.03), suggesting that it captures unique information about firm fundamentals not fully reflected in existing measures.

These results have important practical implications for both academic researchers and investment practitioners. The persistence of CGM's predictive power after controlling for transaction costs and related factors suggests it could be implemented as a viable investment strategy. Moreover, the signal's ability to generate alpha beyond traditional profitability measures indicates that cash generation metrics provide incremental information about firm quality and future performance that warrants inclusion in asset pricing models and portfolio construction frameworks.

However, this study is subject to several limitations that should be acknowledged. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is limited to U.S. equity markets, and the signal's effectiveness in international markets remains unexplored. Second, the sample period and market conditions examined may not fully capture the signal's performance across different economic cycles or market regimes. Third, while we control for several related strategies, the factor zoo is vast, and additional untested factors may subsume some of CGM's predictive power.

Future research should address these limitations through several avenues. International validation of CGM across developed and emerging markets would strengthen the generalizability of our findings. Time-varying analysis could reveal whether CGM's effectiveness varies with market conditions, economic cycles, or changes in accounting standards. Additionally, investigating the economic mechanisms underlying CGM's predictive power—whether it captures operational efficiency, financial flexibility, or market mispricing of cash-generating capabilities—would enhance our theoretical understanding. Finally, examining CGM's performance in combination with other signals through machine learning techniques or optimal portfolio construction methods could yield insights into its role within multi-factor strategies.

Such extensions would not only validate the robustness of our findings but also contribute to the broader understanding of how cash-based metrics inform asset prices in financial markets.

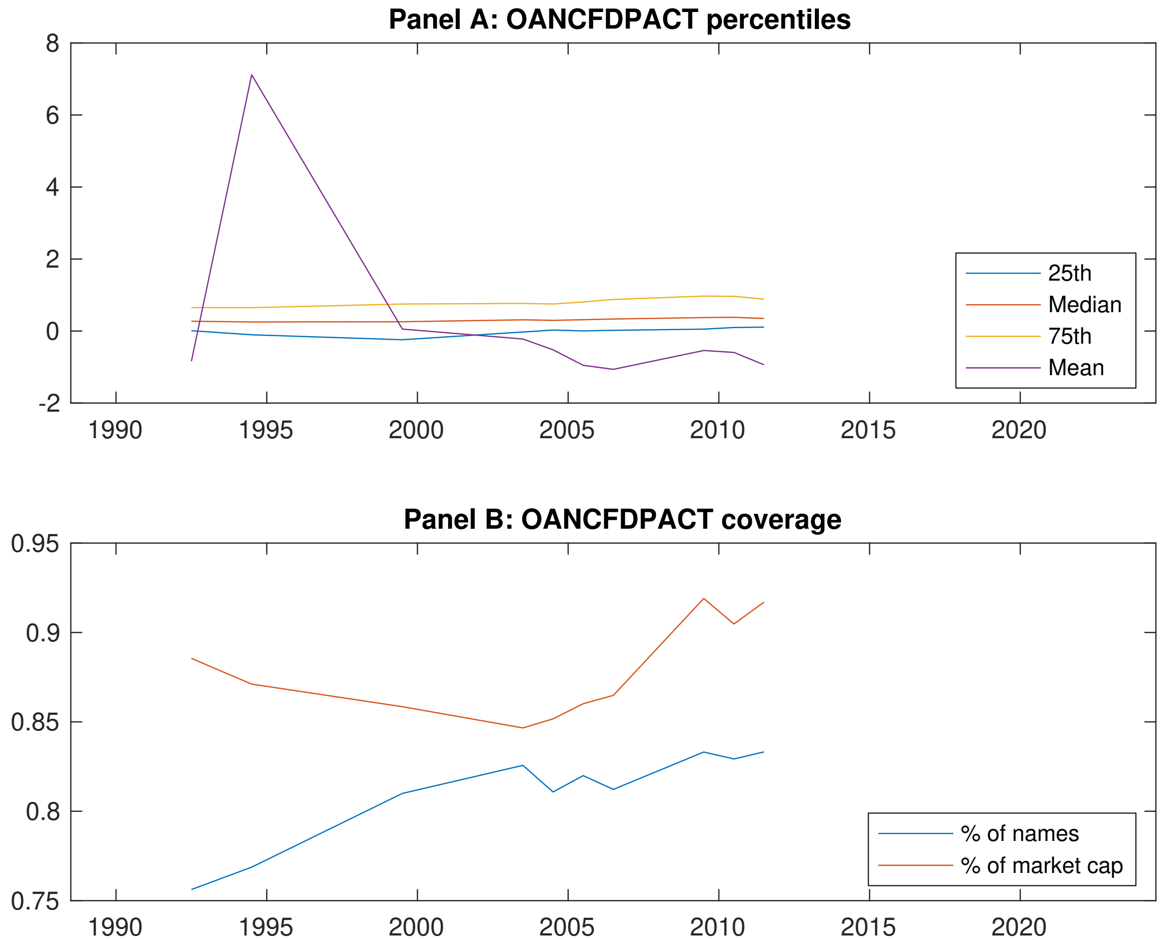


Figure 1: Times series of CGM percentiles and coverage.
This figure plots descriptive statistics for CGM. Panel A shows cross-sectional percentiles of CGM over the sample. Panel B plots the monthly coverage of CGM relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CGM. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015a) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on CGM-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45	0.60	0.74	0.70	0.95	0.50
	[1.53]	[3.13]	[3.52]	[3.50]	[4.03]	[3.17]
α_{CAPM}	-0.49	0.01	0.04	0.02	0.15	0.64
	[-3.80]	[0.14]	[0.54]	[0.34]	[2.22]	[4.06]
α_{FF3}	-0.43	-0.05	0.02	0.01	0.20	0.63
	[-4.43]	[-0.55]	[0.25]	[0.26]	[3.78]	[5.19]
α_{FF4}	-0.39	-0.05	0.03	0.02	0.21	0.60
	[-3.99]	[-0.62]	[0.46]	[0.36]	[3.90]	[4.88]
α_{FF5}	-0.18	-0.21	-0.09	-0.07	0.23	0.41
	[-2.19]	[-2.51]	[-1.26]	[-1.28]	[4.34]	[3.75]
α_{FF6}	-0.16	-0.20	-0.07	-0.06	0.23	0.39
	[-1.88]	[-2.40]	[-0.95]	[-1.12]	[4.34]	[3.51]
Panel B: Fama and French (2018) 6-factor model loadings for CGM-sorted portfolios						
β_{MKT}	1.10	0.88	0.99	0.96	1.05	-0.04
	[52.03]	[41.67]	[56.96]	[73.04]	[79.99]	[-1.54]
β_{SMB}	0.37	0.04	0.08	-0.04	-0.08	-0.45
	[12.24]	[1.33]	[3.05]	[-2.24]	[-4.49]	[-11.51]
β_{HML}	0.01	0.11	-0.02	-0.07	-0.20	-0.22
	[0.37]	[2.91]	[-0.72]	[-3.00]	[-8.79]	[-4.49]
β_{RMW}	-0.56	0.19	0.12	0.17	0.03	0.59
	[-14.72]	[4.88]	[3.78]	[6.99]	[1.05]	[11.77]
β_{CMA}	0.01	0.36	0.25	0.04	-0.17	-0.18
	[0.16]	[6.76]	[5.62]	[1.20]	[-5.19]	[-2.60]
β_{UMD}	-0.04	-0.01	-0.03	-0.01	-0.00	0.04
	[-2.11]	[-0.64]	[-2.14]	[-1.02]	[-0.34]	[1.46]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1373	489	516	582	770	
me ($\$10^6$)	1080	1938	2560	3905	5768	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CGM strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015a) five-factor model, and the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.50	0.64	0.63	0.60	0.41	0.39
			[3.17]	[4.06]	[5.19]	[4.88]	[3.75]	[3.51]
Quintile	NYSE	EW	0.37	0.51	0.43	0.31	0.15	0.05
			[2.03]	[2.83]	[2.76]	[1.98]	[1.00]	[0.32]
Quintile	Name	VW	0.50	0.85	0.71	0.68	0.26	0.25
			[2.03]	[3.67]	[4.21]	[3.95]	[1.76]	[1.67]
Quintile	Cap	VW	0.45	0.32	0.39	0.42	0.53	0.53
			[3.03]	[2.19]	[3.24]	[3.40]	[4.63]	[4.60]
Decile	NYSE	VW	0.59	0.78	0.72	0.70	0.51	0.49
			[3.05]	[4.16]	[4.82]	[4.63]	[3.69]	[3.54]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.48	0.60	0.57	0.56	0.40	0.39
			[3.01]	[3.85]	[4.76]	[4.60]	[3.71]	[3.55]
Quintile	NYSE	EW	0.20	0.33	0.26	0.19		
			[1.10]	[1.78]	[1.59]	[1.17]		
Quintile	Name	VW	0.47	0.82	0.67	0.65	0.27	0.26
			[1.88]	[3.53]	[3.94]	[3.80]	[1.84]	[1.77]
Quintile	Cap	VW	0.43	0.28	0.34	0.36	0.47	0.48
			[2.91]	[1.95]	[2.83]	[2.97]	[4.15]	[4.16]
Decile	NYSE	VW	0.55	0.75	0.67	0.66	0.48	0.47
			[2.86]	[3.98]	[4.49]	[4.40]	[3.49]	[3.41]

Table 3: Conditional sort on size and CGM

This table presents results for conditional double sorts on size and CGM. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CGM. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CGM and short stocks with low CGM .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

		CGM Quintiles					CGM Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.05	0.56	0.89	1.01	0.93	0.88	1.08	0.90	0.85	0.47	0.43
		[0.11]	[1.41]	[2.84]	[3.36]	[3.17]	[3.35]	[4.19]	[4.20]	[3.92]	[2.44]	[2.24]
	(2)	0.32	0.64	0.92	0.87	0.91	0.59	0.89	0.74	0.67	0.31	0.26
		[0.78]	[2.16]	[3.39]	[3.18]	[3.31]	[2.69]	[4.31]	[4.39]	[3.90]	[2.11]	[1.75]
	(3)	0.38	0.79	0.82	0.92	0.94	0.56	0.74	0.69	0.59	0.41	0.34
		[1.10]	[3.13]	[3.17]	[3.53]	[3.36]	[3.41]	[4.61]	[4.44]	[3.80]	[2.82]	[2.32]
	(4)	0.56	0.89	0.75	0.86	0.98	0.43	0.47	0.49	0.42	0.40	0.33
		[1.92]	[3.92]	[3.03]	[3.65]	[3.63]	[3.03]	[3.28]	[3.43]	[2.90]	[2.77]	[2.33]
	(5)	0.44	0.70	0.66	0.92	0.98	0.54	0.34	0.44	0.46	0.59	0.59
		[2.10]	[3.61]	[3.41]	[4.36]	[3.67]	[3.09]	[2.01]	[3.00]	[3.12]	[4.14]	[4.10]

Panel B: Portfolio average number of firms and market capitalization

		CGM Quintiles					CGM Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	399	401	408	413	412	37	34	43	54	58
	(2)	123	123	124	123	123	79	83	86	86	87
	(3)	83	84	84	83	83	139	145	147	148	147
	(4)	70	70	70	70	70	305	320	317	326	329
	(5)	62	62	62	62	62	1774	2119	2410	3265	2713

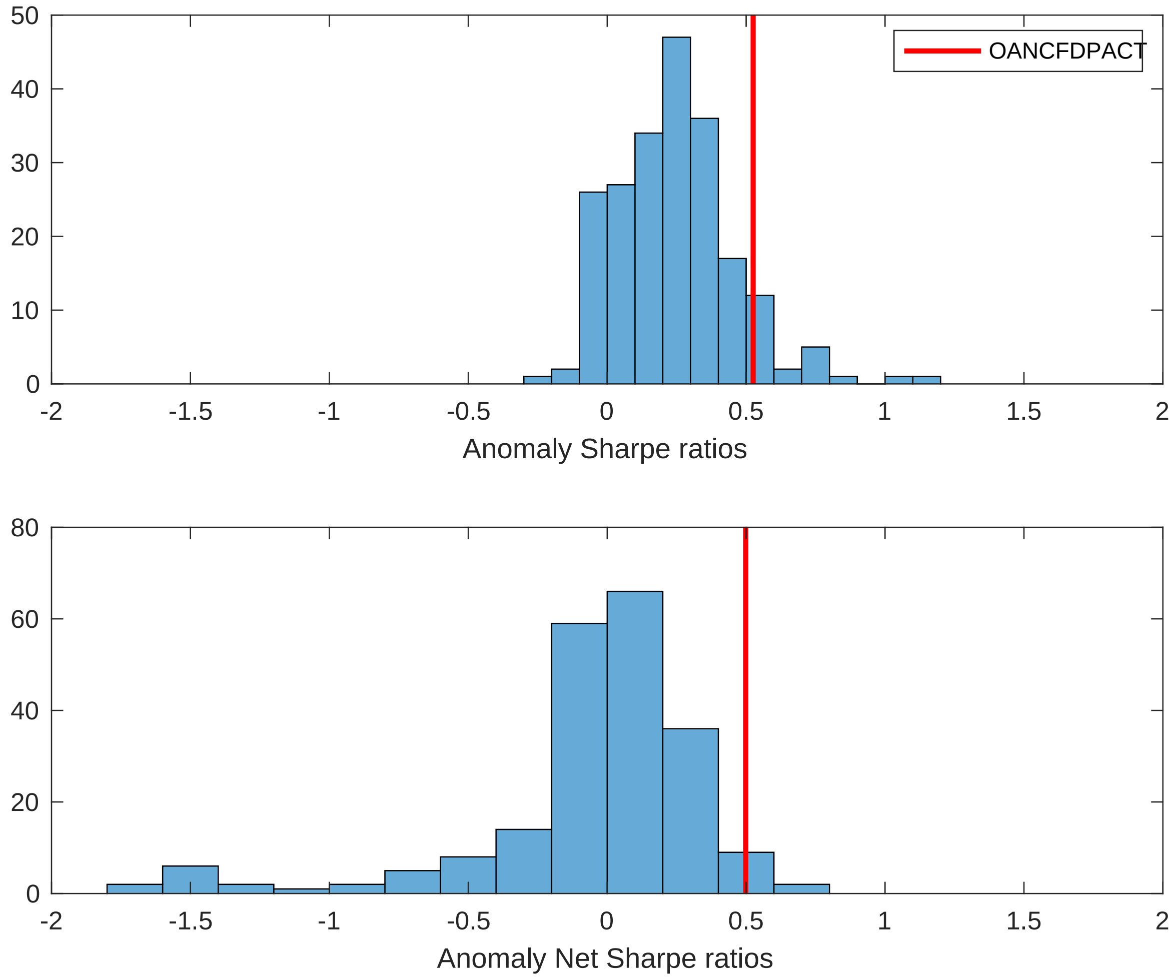


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CGM with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

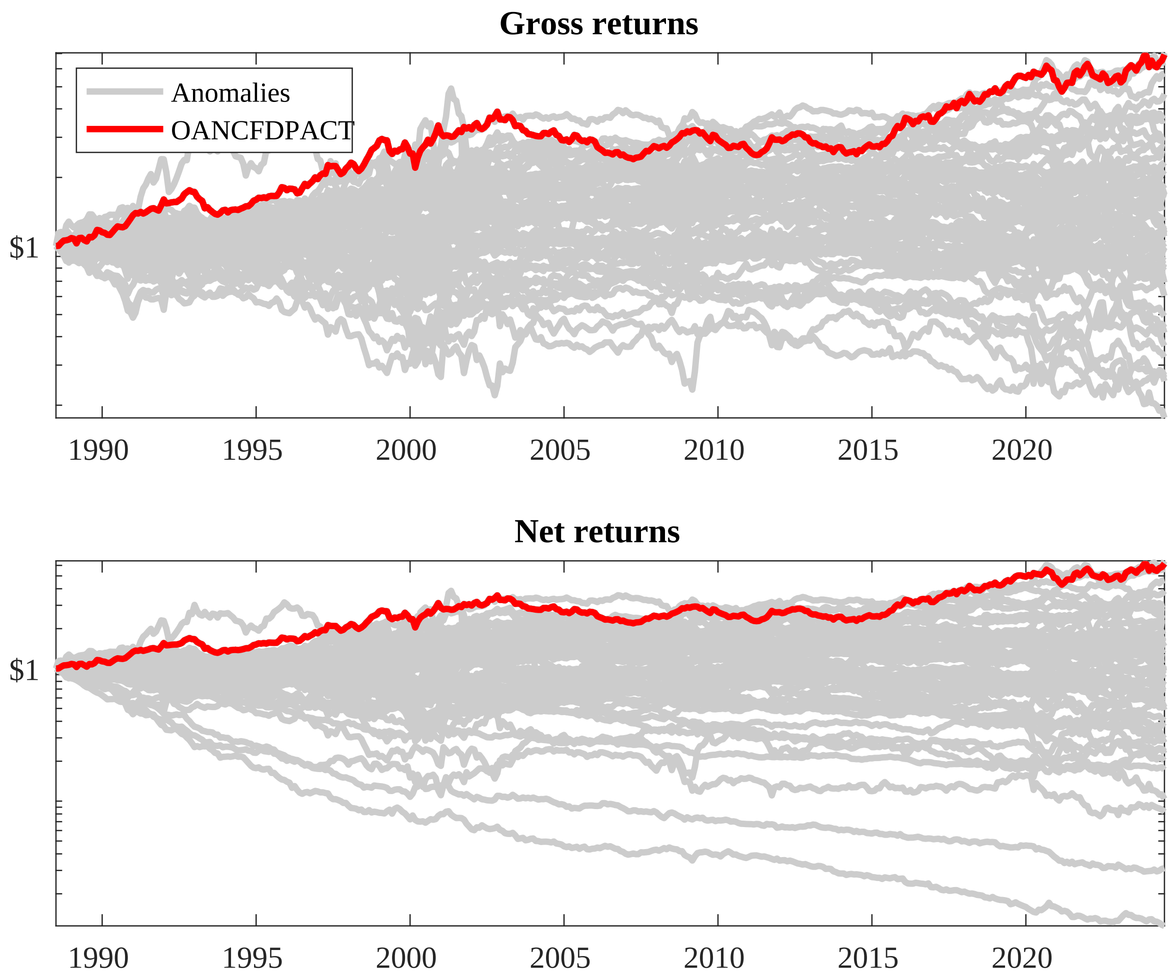
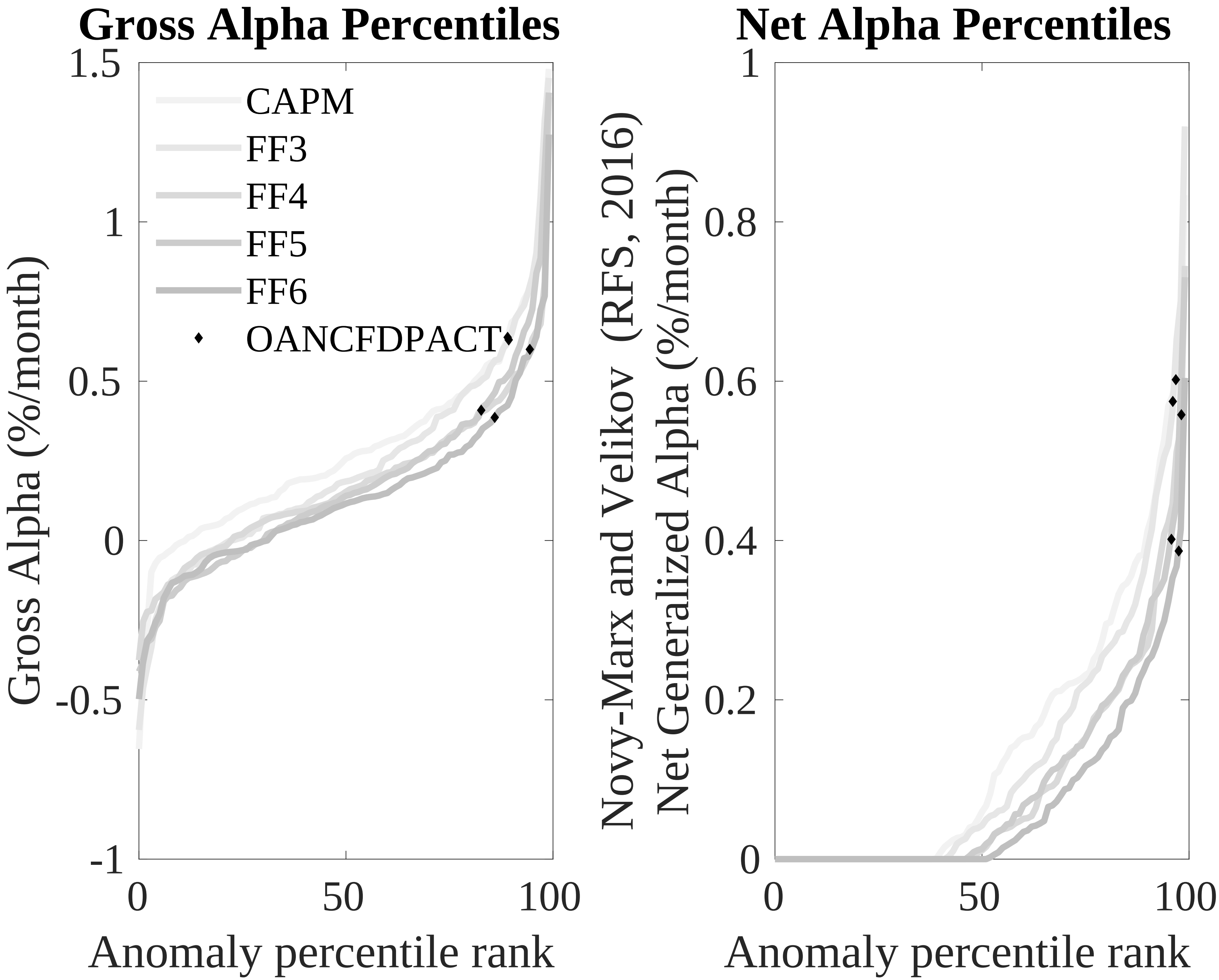


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CGM trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
OANCFDPACT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
1
0.8
0.6
0.4
0.2
0
0
50
100
0
50
100

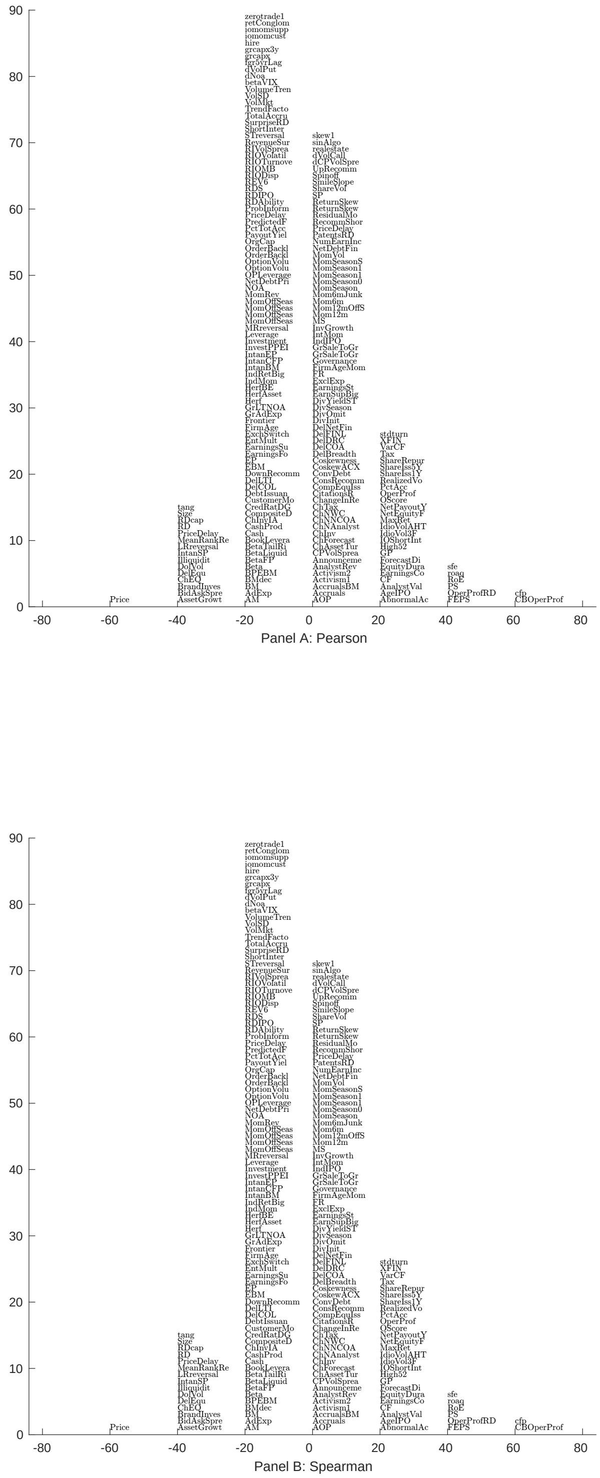


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with CGM. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

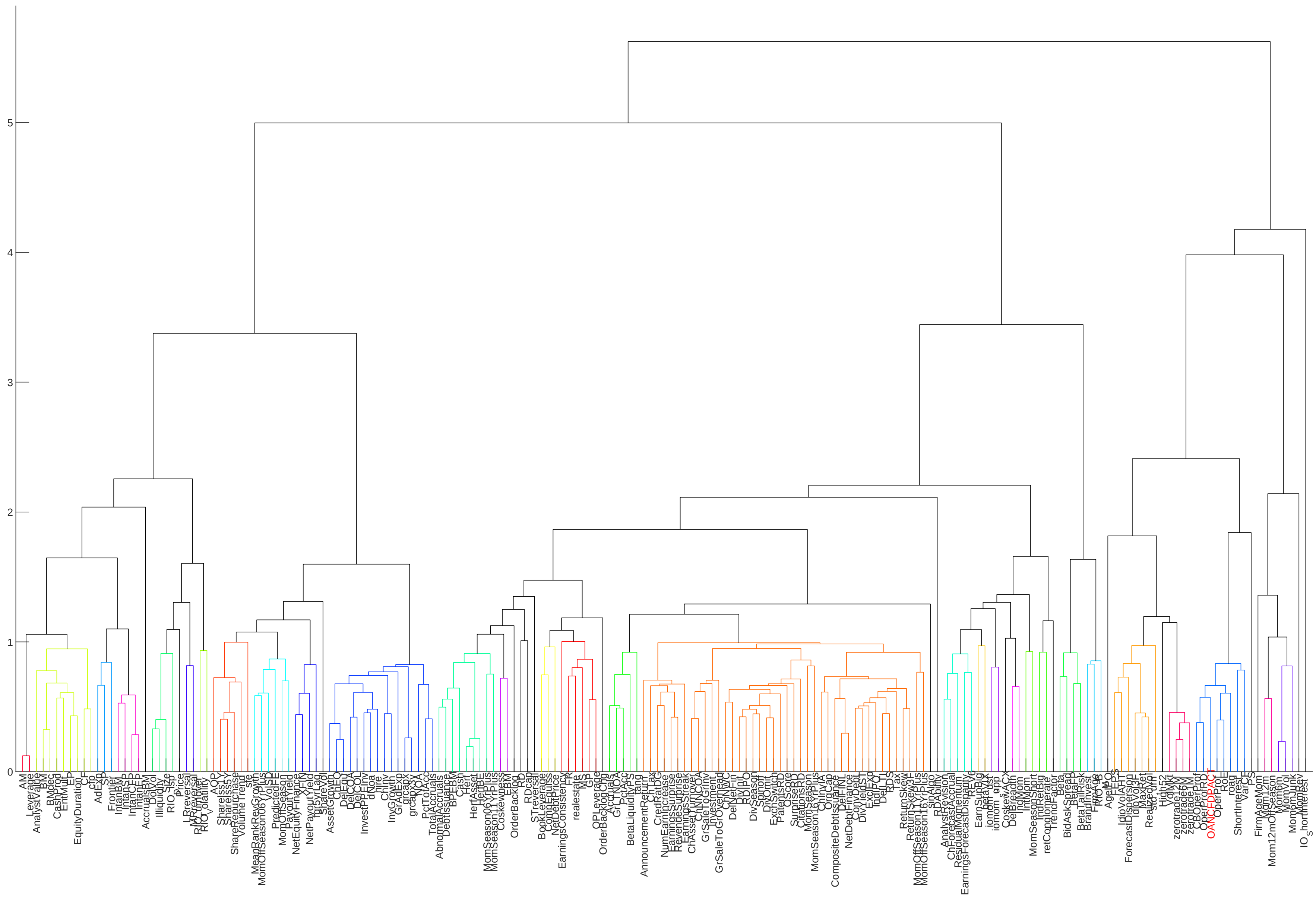


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

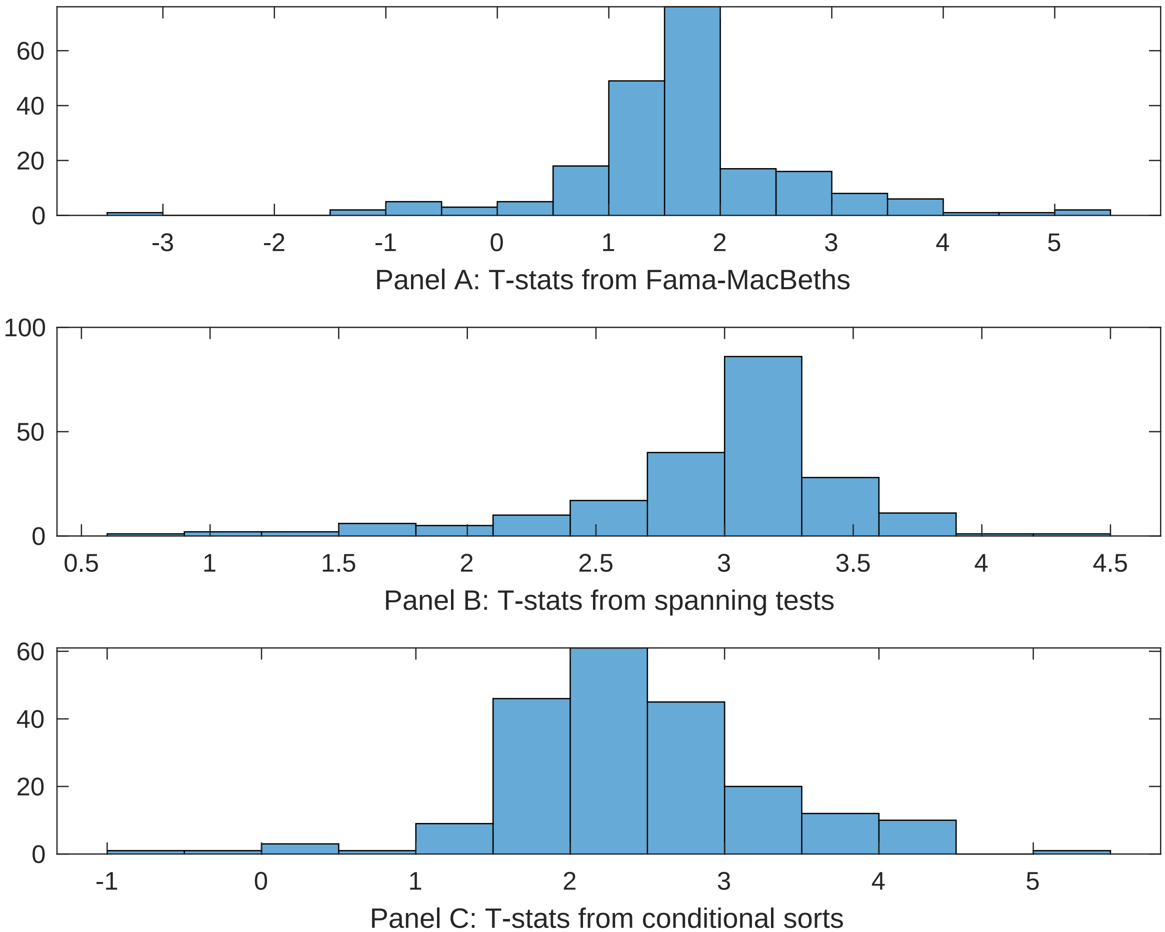


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CGM conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CGM} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CGM} CGM_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CGM,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CGM. Stocks are finally grouped into five CGM portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CGM trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on CGM. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CGM} CGM_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.94	0.91	0.11	0.12	0.12	0.93	0.85
	[2.98]	[2.74]	[3.72]	[4.09]	[4.15]	[2.70]	[2.66]
CGM	-0.63	0.87	0.17	0.54	0.18	0.26	0.24
	[-0.70]	[1.26]	[1.85]	[0.96]	[1.67]	[3.43]	[0.22]
Anomaly 1	0.20						0.15
	[5.03]						[2.97]
Anomaly 2		0.17					0.30
		[3.58]					[0.50]
Anomaly 3			0.56				-0.40
			[0.95]				[-0.48]
Anomaly 4				0.44			0.24
				[2.82]			[1.68]
Anomaly 5					0.12		0.31
					[0.54]		[1.27]
Anomaly 6						0.40	-0.28
						[0.58]	[-0.53]
# months	427	427	432	427	427	427	427
$\bar{R}^2$(%)	1	1	1	1	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the CGM trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CGM} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015a) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Cash-based operating profitability, Operating profitability RD adjusted, net income / book equity, Return on assets (qtrly), Cash-flow to price variance, Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.21	0.22	0.36	0.33	0.34	0.37	0.19
	[2.07]	[2.07]	[3.36]	[3.03]	[3.32]	[3.32]	[2.03]
Anomaly 1	45.59						32.75
	[10.48]						[5.31]
Anomaly 2		35.25					5.46
		[8.30]					[0.87]
Anomaly 3			35.50				8.87
			[5.48]				[1.35]
Anomaly 4				23.91			11.28
				[5.01]			[2.41]
Anomaly 5					31.12		21.51
					[7.79]		[5.59]
Anomaly 6						2.66	-10.73
						[0.63]	[-2.67]
mkt	0.02	1.87	1.12	-0.34	4.90	-3.13	6.25
	[0.01]	[0.71]	[0.40]	[-0.12]	[1.75]	[-1.08]	[2.36]
smb	-32.62	-34.21	-32.61	-38.84	-23.46	-42.01	-21.72
	[-8.88]	[-8.92]	[-7.51]	[-9.81]	[-5.20]	[-8.93]	[-4.73]
hml	-2.73	-6.09	-17.13	-12.68	-6.67	-20.94	10.36
	[-0.59]	[-1.27]	[-3.64]	[-2.57]	[-1.38]	[-4.31]	[2.13]
rmw	35.95	32.86	27.80	37.22	44.23	55.99	22.10
	[7.26]	[5.88]	[3.75]	[5.72]	[8.79]	[8.22]	[3.06]
cma	-27.66	-20.82	-14.99	-20.26	-9.05	-19.69	-15.09
	[-4.39]	[-3.20]	[-2.21]	[-2.97]	[-1.36]	[-2.76]	[-2.40]
umd	-0.08	-0.88	1.73	-2.38	-2.22	3.17	-4.89
	[-0.04]	[-0.38]	[0.73]	[-0.89]	[-0.92]	[1.22]	[-2.02]
# months	428	428	432	428	428	428	428
$\bar{R}^2$(%)	66	63	60	59	62	57	69

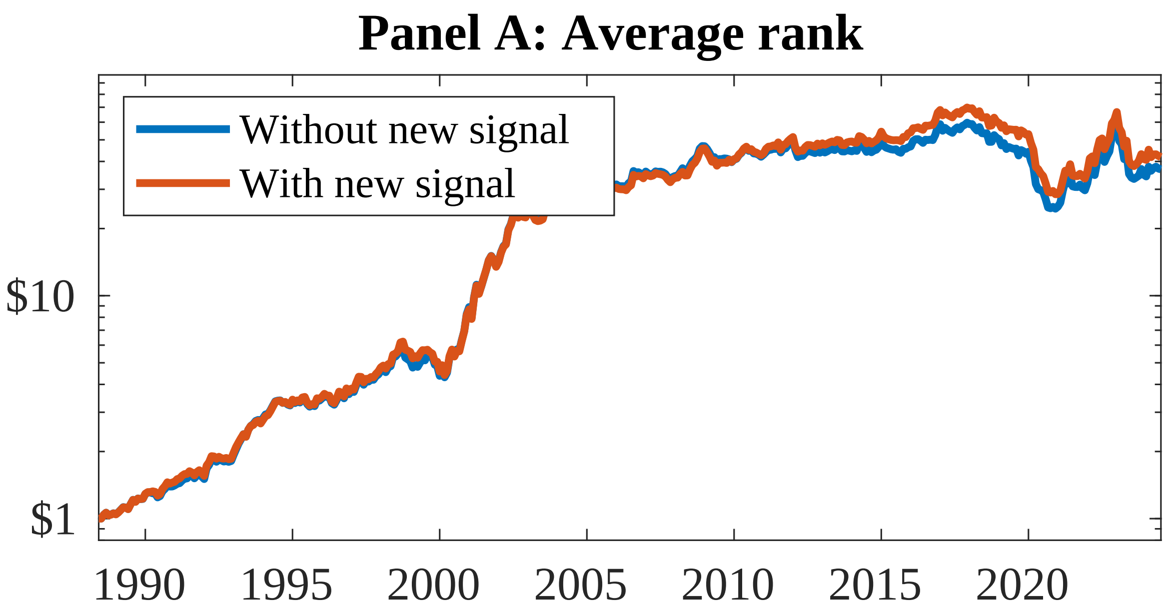


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as CGM. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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